

Influence of enrichment material and herbal compounds in the behaviour and performance of growing pigs



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ABSTRACT

Pigs reared in barren conditions are exposed to many different stressors, compromising their welfare and producing physiological and behavioural changes. The aim of this study was to assess the effect of environmental enrichment (EE) consisting of natural hemp ropes, sawdust, rubber balls, and a herbal compound (HC) of *Valeriana officinalis* and *Passiflora incarnata* on the behaviour and performance of growing pigs. Fifty-six pigs were used to assess four different treatments divided in two pens of seven animals per treatment (14 pigs/treatment). The treatments tested were: (a) pigs reared with EE, (b) pigs supplemented with HC, (c) pigs provided with both EE and HC, and (d) control group (CG, neither EE nor HC). Body weight and lesions were measured before starting the experiments (week 15) and at 18, 20, 22 and 24 weeks of age. Weekly instantaneous scan and continuous focal sampling were used to record behavioural patterns of activity, social interactions and abnormal behaviours. Three novel tests were carried out at 16, 19 and 23 weeks of age. Body weight at the end of the experiment was found to be significantly lower for the pigs reared in the control group compared to the other treatments ($p = 0.0009$). Furthermore, pigs reared with EE presented less stereotypies ($p = 0.016$) and redirected behaviour ($p = 0.0188$), but more exploratory behaviour ($p = 0.008$). However, pigs supplemented with HC presented less social interactions ($p = 0.048$), a trend to present less negative social behaviour ($p = 0.09$) and less skin lesions ($P = 0.0433$) than pigs not supplemented. Finally, no remarkable differences were reported in any of the three novel tests. Thus, both EE and HC positively influenced some animal welfare indicators and performance of growing pigs in the present experiment.

1. Introduction

Most of the European pig production is based on intensive farming. Intensification mainly aims at optimising production costs, but it may present some detrimental implications on animal welfare grounds. The interest in animal welfare has globally increased (Sandoe and Simonsen, 1992), driving a higher demand for more “animal friendly” systems. Although the concept of animal welfare allows different interpretations (Mason and Mendl, 1993), it is generally accepted that stress negatively influences animal welfare (Veissier and Boissy, 2007). Stress is often defined as a threat to the homeostasis of the organism (Chrousos and Gold, 1992) that activates a broad range of complex physiological, behavioural and neurological changes (Chrousos, 2009). If this response is prolonged, inadequate, excessive, or there is a failure to cope with the stressor it may produce adverse consequences on the physiology of the organism producing immunological, metabolic, reproductive and behavioural alterations (Möstl and Palme, 2002).

Growing pigs in barren environments are exposed to many stressful

factors, with the subsequent negative implications for their welfare and health (Ruis et al., 1997). In fact, a barren environment has been suggested as an important stress factor by itself because pigs present limitations to express their foraging species-specific behaviour, leading to frustration (for a review see: van de Weerd and Day, 2009). Environmental enrichment motivates the exploratory behaviour, which is considered a need for pig's welfare (Studnitz et al., 2007), thus, to provide enrichment material for pigs is one of the strategies to reduce stress.

Another strategy to reduce stress is by means of plants with sedative and tranquilizer properties such as *Valeriana officinalis* and *Passiflora incarnata* (Murphy et al., 2010; Peeters et al., 2004; Soulimani et al., 1997). The mechanism involved in the sedative and anxiolytic properties of *Valeriana officinalis* seems to be mediated by the interaction of Valerian Acid with the γ -Amino butyric acid receptors type A (GABA_A). The stimulation of GABA_A opens the permeability of chloride channels, producing neural inhibition (Khom et al., 2007; Murphy et al., 2010). Moreover, different bioactive compounds have been detected in

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Passiflora incarnata such as flavonoids, maltol, cynogenic glycosides and indole alkaloids, without a consensus regarding the most important one for the sedative properties. Flavonoids, which are beyond the most studied components, have a similar effect to Valerian Acid, increasing the membrane permeability by means of the modulation of GABA_A (for a review see: Miroddi et al., 2013). However, up to the knowledge of the authors, the effect of herbal compounds on welfare in pigs has only been published twice (Peeters et al., 2004, 2006).

Novel object tests are commonly used to assess fear and anxiety (Dalmat et al., 2009), although results are not always consistent and easy to interpret. A higher latency to touch a novel object and less time spent investigating the stimulus have been related with higher levels of anxiety or fear (Murphy et al., 2014). On the other hand, an increased excitability, a quicker approach and longer interactions in front of a novel object have been described in pigs reared in barren conditions compared with pigs reared in enriched conditions (Bracke and Spooler, 2008; Grandin et al., 1987; Pearce and Paterson, 1993; Stolba and Wood-Gush, 1980).

The aim of this work was to study the effect of environmental enrichment by means of natural hemp ropes, sawdust and rubber balls, and the effect of supplementing the diet with a herbal compound containing Valerian (*Valeriana officinalis*) and May pop (*Passiflora incarnata*) on the behaviour at group level or when confronted to novelty and on other animal-based measures (body weight and skin lesions) in growing pigs. Our hypothesis was that pigs raised in an enriched environment and/or supplemented with these herbal compounds should be less stressed and present less abnormal behaviour, less skin lesions and would grow faster than pigs kept in barren conditions and/or without this supplement in the diet.

2. Materials and methods

2.1. Animals and housing conditions

Fifty-six entire males [(Landrace × Large white) × Pietrain] were used in this experiment which lasted 9 weeks (from 16 to 24 weeks of age). Pigs arrived to the facilities at the age of ten weeks with a mean weight of 25 ± 2.04 kg, and were housed together before starting the experiment. At the age of 16 weeks and a mean weight of 49.8 ± 4.27 kg, pigs were randomly distributed between two different rooms. Each room had four pens of 13.67 m^2 (two on the left and two on the right separated by a corridor), with spindles as walls and partly slatted floor. The stocking density per pen was $1.95 \text{ m}^2/\text{pig}$. Food and water were provided *ad libitum* by two hoppers and a single nipple drinker. This distribution allowed to test four treatments in two replicates. Since the aim of this experiment was to study the effect of environmental enrichment and supplementation of an herbal compound, the treatments assessed were as follows: (1) pigs supplemented with both environmental enrichment and herbal compound (named as EEHC pigs from this point), (2) pigs supplemented with environmental enrichment (EE), (3) pigs supplemented with the herbal compound (HC) and (4) control group (CG) consisting of pigs kept in a barren environment and without herbal compound supplementation. The enrichment material consisted of a mixture of two point-source objects (i.e. devices/objects often limited in size and restricted to a single location), natural hemp ropes and rubber balls, in this case; and sawdust, all provided at the same time and during all the experiment. In order to support the sawdust, 1/3 of the slats in the enriched pens were partially covered with polypropylene sticks (Click-in®, Rotenca, Agramunt, Spain), and new sawdust was added every two days. A single rubber ball with a diameter of 15 cm was provided in each enriched pen. Two hemp ropes were hung in the walls of the enriched pens, and were substituted when the length of the rope was shorter than 30 cm. The herbal compound used (Sedafit ESC, Phytosynthèse, Saint-Bonnet de Rochefort, France) contained Valerian (*Valeriana officinalis*) and Maypop (*Passiflora incarnata*) and it was manually added to the food

concentrate (2000 mg/kg).

Weight was recorded at the age of 15 (before starting the experiment), 18, 20, 22 and 24 weeks using a cage with a scale (MBWA100 Meier-Brakenberg; GmbH & Co, Germany). Skin lesions were assessed at 15 weeks of age (baseline level), the day after mixing the animals and at 18, 20, 22 and 24 weeks of age taking advantage of the restraint provided by the cage during the weighing procedure. The total amount of lesions in one side of the pig in each one of the five regions (ears, front, middle, hind-quarters and legs) defined in the Welfare Quality® (2009) were recorded, in addition to scoring the animal as 0, 1 and 2 according to the protocol. Furthermore, blood, saliva and hair samples were taken before starting the experiment (15 weeks of age), at the middle (20 weeks of age) and at the end (24 weeks of age) for cortisol, chromogranin A and TNF- α quantification as part of a broader study. The results from physiological indicators have been published elsewhere (Casal et al., *in press*) and will not be reported in this paper. All the procedures carried during this experiment were approved by the IRTA ethical committee.

2.2. Behavioural observations

Instantaneous scan sampling and continuous focal sampling as described in Martin and Bateson (1993) were used to record behaviour. Once per week, during nine weeks, direct observations of each room (4 pens) in sessions of two consecutive hours (11:00–13:00) were carried out by a trained observer. The observations were performed in a specific order to avoid possible differences of time between groups. Scan samplings were taken every ten minute intervals, and focal samplings elapsing six minutes per pen were recorded between two consecutive scan samplings (i.e. instantaneous scan samplings of activity were performed for 1 min at the beginning and end of each observation round and 6 min of focal sampling was performed in between). Thus, each observation day provided a total of 12 scans per animal and 18 min of focal sampling per pen, divided in three periods of six minutes. The behaviours observed according to each observational methodology are summarised in Table 1. A new category named “Exploratory behaviour” was created by grouping the categories “interaction with the pen” and “interaction with the enrichment material”. Twenty minutes before starting the observations, the observer entered into the room and walked for ten minutes among the corridor with the aim of allowing the pigs to get used to his presence. Then, the observer moved to the centre of the room, and stayed there for another ten minutes before starting the observations. Observations were done from the centre of the room with full visual contact to all the pens.

2.3. Novel object tests

At 16 weeks of age pigs were subjected to a novel object test in a novel environment. The novel object was a traffic cone filled with cement to avoid the potential movement produced by pigs when contacting it. A second novel test was performed at 19 weeks of age, and consisted in a PVC ball of 55 cm (Fitball 55 cm Domyos, Decathlon, Villeneuve-d'Ascq, France) suspended from the ceiling at 30 cm from the floor. The pigs did not have previous experience with a hanging ball, as the enrichment material provided was a rubber ball freely available on the floor of different colour, size and texture. Finally, a third novel test was performed with a human shape at 23 weeks of age; the test consisted in a caretaker dressed in a white overall with cowl and with a disposable mask. All the novel tests were performed in the same room. The test room (Fig. 1) was 31.5 m^2 ($4.7 \text{ m} \times 6.7 \text{ m}$), and was at a distance of 6.4 m from one of the rooms and at 20.6 m of the other room. In order to balance the distance, a short walk was done with the animals of both treatments before starting the test so that the final time taken to reach the test pen was the same (close to one minute). All pigs were conducted to the test room by the same caretaker. The novel objects were placed straight to the entrance at 3.5 m.

Table 1
Ethogram used for both scan and focal samplings.

Specific category	Definition
SCAN SAMPLING	
Activity	
Standing inactive	Pig is upright on all four legs, neither moving forward or backward
Walking	Pig is upright on all four legs, and moves in the pen
Sitting inactive	Pig is upright on two front legs, and hindquarter
Lying	Pig is recumbent on its belly or side
Behaviour	
Eat/Drink	Head or snout over bowl or trough
Interaction with the pen ^a	Licking, chewing, nosing or sniffing unanimated objects from the pen, excluding enriched material
Interaction with enrichment material ^a	Interaction with sawdust, ropes or balls
Social Interaction	Head or snout in contact with another pig (includes any positive or negative social behaviour)
Other behaviours	Behaviours different than those previously described
FOCAL SAMPLING	
Positive social behaviour	Head or snout in contact with another pig. The receiver doesn't react negatively
Stereotypies	Stereotypic behaviour, basically animal rubs the bars of the crate with the mouth, and other stereotypic behaviours
Redirected behaviour	Vertical rub movements with the snout to the belly of a pen-mate (belly nosing), bite or suck the ear of a pen-mate (ear biting) and bite or suck the tail of a pen-mate (tail biting)
Negative social behaviour	Aggression, fights to another pig with a negative response from the receiver
Sexual behaviour	Sexual behaviour, one animal mounting or trying to mount another pig
Other behaviours	Any other behaviour

^a Exploratory behaviour was analyzed grouping the Interaction with the pen and the interaction with the enrichment material (EM).

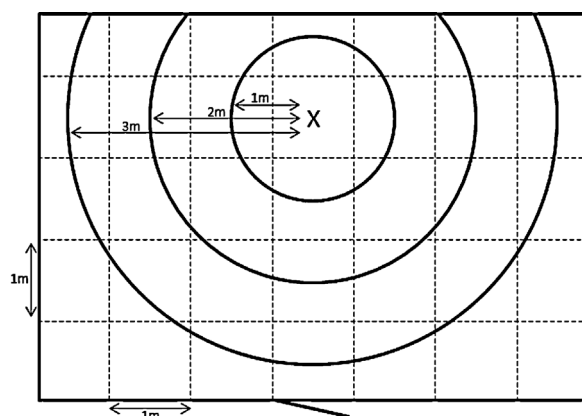


Fig. 1. Schematic representation of the novel test room with concentric lines every meter and lines drawn in the floor for activity evaluation. X represents the place where the object was placed.

Linear and concentric lines were drawn on the floor every meter for activity analysis, so that the arena test was divided in 4 areas by 3 concentric lines from the novel object (1m, 2 m and 3 m lines). Vertical and horizontal lines divided also the arena floor in squares measuring 1 m². Observations for behaviour analyses were done for up to five minutes through a door window. After five minutes, if the animal did not touch the object, it was conducted to its home pen. All the observations were carried out by the same observer. The parameters observed in the novel test were time to cross the concentric line of two and one meter to the object, latency to contact the object, total lines crossed, number of extra contacts, number of vocalizations, reluctance to enter into the room and excretory behaviour.

2.4. Statistical analysis

All the statistical analyses were conducted using Statistical Analysis System (SAS version 9.2; SAS institute Inc., Cary, NC; USA). Significance was established at $p < 0.05$ for all the analyses, while a tendency was considered between $p > 0.05$ and $p < 0.1$. Results are presented as mean $\pm$ standard error (SEM) unless otherwise indicated. Shapiro-wilk test (with PROC UNIVARIATE in SAS) was used to examine the normality of distributions. The experimental unit for weight, lesions and novel test was individual, whereas for behavioural

observations pen was considered as experimental unit. Least square means of fixed effects (Lsmeans) with tukey adjustment were used for comparisons when analysis of variance indicated differences ($p < 0.05$).

Weight presented a parametric distribution, and it was analysed using generalized linear mixed model (MIXED procedure and the covariance matrix AR(1) in SAS) for repeated measurements. Use or not of herbal compound and environmental enrichment, sampled day and their interactions were considered fixed effects, while pen was considered as a random effect. Furthermore, weight before starting the experiment was considered as a covariate after observing no differences between treatments.

Skin lesions and novel tests were analysed by means of non-parametric generalized linear models (GLIMMIX or GENMOD procedure, respectively). For skin lesions a negative binomial distribution was used and for novel tests either a negative binomial or a Poisson depending on the deviance (Cameron and Trivedi, 1998). Use or not of enrichment material and herbal compounds and the interaction were considered as fixed effects. For skin lesions, sample day was also considered as fixed effect, lesions before starting the experiment was used as a covariate and pen was included as a random effect.

Behavioural data from scan samplings were analysed as a percentage of the scans of each category in relation with the total number of scans per day, while data from focal sampling were analysed as the total number of behaviours observed in each category per pen and day. A first analysis of behavioural data was carried out with a non-parametric generalized linear model (GENMOD procedure) with repeated measurements and the covariance matrix AR(1). However, no effect of time was observed on any of the parameters, and it was removed from the models for simplicity, comparing only the mean values of all observations throughout time by means of another non-parametric generalized linear model (GLIMMIX procedure). A Poisson or a negative binomial distribution was used based on the deviance (Cameron and Trivedi, 1998). Use or not of enrichment material and herbal compound and their interaction were considered as fixed effects and pen as a random effect. For behavioural data related to activity an interaction was found between EE and HC supplementation. Since experimental unit for behavioural data was pen, the results for these activity observations come from comparisons of two pens per treatment and not from four as when contrasting per main factors (EE and HC) like for the rest of behavioural parameters presented in which no interaction was found. For that reason, the authors present only descriptive statistics of these activity

Table 2
Live weight of pigs of the different treatments throughout the experimental time.

n	ALL	CG	HC	EE	EE HC	SEM
	55	14	14	14	13	
18 weeks	64.82	64.46	65.24	65.49	64.09	0.69
20 weeks	79.02	77.64	78.28	79.27	80.86	0.69
22 weeks	91.94	89.43 ^a	90.78 ^a	93.63 ^b	93.90 ^b	0.69
24 weeks	109.82	104.5 ^a	110.35 ^b	111.79 ^b	112.35 ^b	0.69

SEM = standard error of least square means.

Least Square means in a row with different superscripts differ significantly at $p < 0.05$. EM = pigs grown in an enriched environment, HC = pigs supplemented with a herbal compound with sedative properties, EM HC = pigs grown in an enriched environment and supplemented with herbal compound, and CG = pigs with neither enrichment material nor herbal compound (control group).

parameters.

3. Results

3.1. Body weight

Body weight at 16 weeks was not significantly different between treatments (50.22 ± 1.17 ; 49.61 ± 1.05 ; 49.64 ± 1.26 ; 49.81 ± 1.22 , for CG, HC, EE and EEHC, respectively) and was used as a covariable. Table 2 summarises the results of the interaction between treatments throughout time ($F = 2.46$; D.F = 15/308; $p = 0.0009$). Pigs in the control group presented a significantly lower weight at 24 weeks of age compared to EE, HC and EEHC pigs.

3.2. Body lesions

Less body lesions were observed in the groups of pigs supplemented with the HC compared with those not supplemented (mean body lesions per day: 5.32 ± 0.41 vs 7.12 ± 0.55 respectively; $F = 4.15$; D.F = 1/152; $p = 0.043$).

3.3. Behavioural observations

No time effect was found on any of the behavioural parameters recorded, thus, time was removed to simplify comparisons between treatments. With regards to activity patterns, EE pigs stayed less time lying or sitting compared with CG, HC and EEHC, although no statistics have been carried out for that particular parameter as explained in Section 2.4. (Fig. 2). The effect of environmental enrichment and supplementation of herbal compound on the other behaviours recorded is

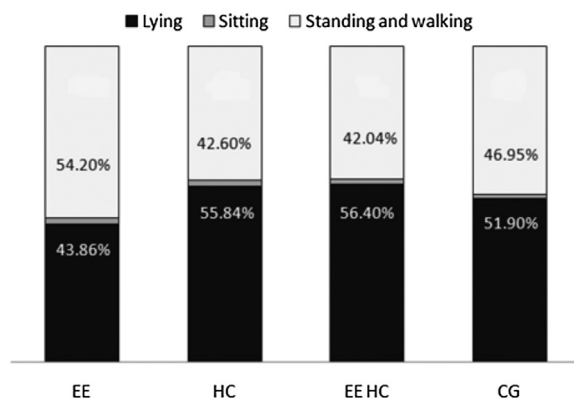


Fig. 2. Percentage of activity patterns of pigs subjected to the different treatments: pigs grown in an enriched environment (EE), pigs supplemented with a herbal compound with sedative properties (HC), pigs grown in an enriched environment and supplemented with herbal compound (EEHC), and pigs with neither enrichment material nor herbal compound (CG).

Table 3
Effect of environmental enrichment and herbal compound supplementation on behaviour of pigs during the whole study.

Behaviour	Housing system ^b		Herbal compound ^c		p-value		
	BE	EE	no	yes	SEM ^a	HS ^b	HC ^c
Scans							
Exploration	4.17	12.13	8.87	7.42	0.49	**	ns
EM ^d interaction	–	–	11.42	7.22	0.45	–	ns
Social interactions	4.06	3.85	5.23	2.68	0.25	ns	*
Focals							
Positive social behaviour	5.84	4.31	4.56	5.59	0.39	0.09	ns
Stereotypies	4.88	1.22	2.97	3.13	0.43	*	ns
Redirected behaviour	0.75	0.28	0.50	0.53	0.09	*	ns
Negative social behaviour	7.25	5.84	7.84	5.25	0.77	ns	0.1
Sexual behaviour	0.38	1.19	0.78	0.78	0.17	0.06	ns

p-values are represented by * $P < 0.05$; ** $p < 0.01$; *** $p < 0.001$; tendency $p < 0.1$ and ns not significant. Behaviours from scan samplings are expressed as a% of total scans per pen while behaviours from focal samplings are expressed as mean occurrence of the behaviours per pen. Time was removed to simplify comparisons between treatments. Interaction HS*HC was not significant for any parameter.

^a SME: Standard error of the mean.

^b Housing system (HS) BE = Barren environment/EE = enriched environment.

^c Herbal compound (HC) yes = supplied with herbal compound/no = not supplied.

^d EM = Enrichment material consisting in sawdust, ropes and balls.

presented in Table 3. Pigs reared in enriched conditions performed less stereotypies ($F = 10.17$; D.F = 1/5; $p = 0.016$) and less redirected behaviour ($F = 7.73$; D.F = 1/5; $p = 0.0188$) than pigs reared with no EE. Furthermore, enriched pigs presented more exploratory behaviour compared with no enriched pigs ($F = 12.04$; D.F = 1/5; $p = 0.008$). Furthermore, enriched pigs tended to have less positive interactions than no enriched pigs ($F = 2.64$; D.F = 1/5; $p = 0.089$), and showed a tendency to present more sexual behaviour ($F = 3.16$; D.F = 1/5; $p = 0.06$). On the other hand, animals supplemented with the herbal compound showed less social interactions ($F = 5.10$; D.F = 1/5; $p = 0.048$) than pigs not supplemented, although no significant differences but a trend was reported during the focal samplings, with lower negative social behaviours ($F = 2.40$; D.F = 1/5; $p = 0.09$) pigs supplemented with HC compared with those not supplemented.

3.4. Novel test

No significant differences were observed in neither the novel test performed with a cone (16 weeks of age), nor the novel test performed with the human shape (23 weeks of age) in any of the parameters observed. When a PVC ball was used (19 weeks of age), pigs with no EE crossed the two-meter concentric line from the novel object before than pigs grown in enriched conditions (Chi-Square = 86.59; D.F = 51; $p = 0.012$). Furthermore, the activity of the pigs during the novel test tended to be higher for the pigs reared with no EE at 16 weeks (Chi-Square = 58.11; D.F = 52; $p = 0.10$) and at 23 weeks of age (Chi-Square = 56.61; D.F = 52; $p = 0.07$) in comparison to enriched pigs.

4. Discussion

This study adds support to the hypotheses that enrichment material and some bioactive constituents from herbal compounds have a positive effect on the welfare and performance of growing pigs.

In the present study pigs supplemented either with enrichment and/or herbal compound were found to present a significant increase in body weight compared with the control group at 22 and 24 weeks of age. This finding could indicate that environmental enrichment and herbal compounds can improve the growth of pigs since they were

slaughtered at the same age. Furthermore, these results are in agreement with the work of Beattie et al. (2000) who described a higher body weight, feed intake, food conversion and growth rate in pigs from a similar age raised in an enriched environment (weights comprised between 15 and 21 weeks of age). Moreover, increased growth rates have been previously related with an increased food intake in pigs reared with straw (Lyons et al., 1995). Although no straw was used in the present experiment, sawdust, which is a similar material even preferred to straw in some studies (Beattie et al., 1998), was provided to the enriched pens together with point-source objects (rubber balls and ropes, as explained previously). Literature on the effects of point-source objects on performance present more inconsistent results compared to the use of straw (Van de Weerd and Day, 2009). Improved animal growth was reported when providing car tires and both mineral block and a teeter-totter device consisting of a metal bar with rubber bells on the ends (Schaefer et al., 1990). However, Hill et al. (1998) found an increased average daily gain and food:gain ratio in the animals provided with point-source enrichment, but only in the finishing stage and for one of the genetics tested (EXP-94). Two main hypotheses are proposed to explain the increased body weight reported in the pens with enrichment material. First hypothesis, when enrichment is provided, pigs are more active and present a higher exploratory behaviour (Lyons et al., 1995), which facilitates feeder visits and, consequently, may increase the feed intake. Indeed, as it is highlighted below, an increase in exploratory behaviour was reported in the present experiment in pigs reared in enriched environments. The second hypothesis of this study, is related with the reduction of stress reported in animals reared in enriched conditions (Schaefer et al., 1990), which have been described to reduce cortisol and catecholamines, both with a catabolic activity (Mormède et al., 2007). With this regards, De Jonge et al. (1996) reported a potential effect of cortisol on feed efficiency and performance. As described in the materials section, physiological indicators of stress were also evaluated in the pigs of the present study, and significant lower levels of cortisol were found in enriched pigs compared to those housed in barren environment (Casal et al., *in press*). This would support the potentially catabolic role of cortisol, which could explain the lower weight of barren housed pigs, although further investigations should explore more on this aspect regarding the trade-off between energy expenditure for activity or for growth.

Body weight was higher also for those pigs supplemented with the herbal compound. No other studies have been found regarding the effects of herbal compounds use on body weight. A reduced number of skin lesions was observed in the animals supplemented with the herbal compound together with a trend for a reduced number of social negative behaviours. Thus, a possible explanation for the higher growth could be that the sedative effects of the herbal compound, in the present study, would be reducing the number of aggressions and probably diverting energy from negative social behaviour to growth. These results are in discordance with those reported by Peeters et al. (2006) who found more lesions in the shoulder and loin of pigs supplemented with the herbal compound, although this finding was not expected by the authors. In both experiments, the product used was the same (Sedafit) but while in Peeters et al. (2006) the herbal compound was dissolved in the water, in the present study it was mixed with the food. Furthermore, in Peeters et al. (2006) the evaluation of the lesions was done at the slaughterhouse line after different acute stressors such as fasting for 18 h, mixing and transporting the pigs to the abattoir, while in the present study lesions were assessed periodically on farm.

On the other hand, environmental enrichment did not have an effect on neither negative interactions nor skin lesions. Although some studies reported reduced levels of aggression in enriched environments, literature shows no general agreement on results (Van de Weerd and Day, 2009). The increased amount of aggression in barren pens can be a consequence of the increased levels of redirected behaviour to the pen mates (Beattie et al., 1996). Nevertheless, deficient enrichment materials can increase the aggressions by enhancing the competition, which

can produce more negative agonistic behaviour, producing frustration and/or aggression (Docking et al., 2008). Furthermore, more aggression has also been previously related with a bad location of the enrichment material, since it can cause disruption to other pen mates (Van de Weerd and Day, 2009). Thus, type and quality of the material, location, novelty effect, and number of objects should be taken into account so that the enrichment becomes meaningful and available to pigs, properly enhances exploratory behaviour and obtains the expected beneficial effect on their welfare. Together with that, the biosecurity and/or sanitary issues and potential challenges for manure management systems associated with materials like sawdust, which may increase dustiness, have to be taken into account.

The present experiment also observed a significant increase in the exploratory behaviour of pigs reared in enriched environments compared with pigs reared in barren environments in agreement with previous studies (Beattie et al., 2000). Pigs reared with environmental enrichment have been said to be more stimulated (Van De Weerd et al., 2003). The manipulation of the substrates provided as enrichment better fulfils the pigs' exploratory needs, since pigs have a preference for chewable, destructible, rootable and deformable materials (Bolhuis et al., 2005). The present study confirms that properly manipulative materials are those capable of increasing exploratory behaviour in pigs, since no significant differences in exploration were found when only considering interactions with pen structures. Exploratory behaviour is considered as a need for pig's welfare; as a consequence, pigs reared in barren conditions are more susceptible to present behavioural problems such as stereotypies and redirected behaviours to other pen mates in an attempt to satisfy their exploratory needs (Studnitz et al., 2007). Stereotypies are considered a repetitive, invariable and functionless behaviour closely related with stress and poor animal welfare (Mason et al., 2007). The present study reported less stereotypies in pigs reared in enriched conditions compared with pigs reared in barren conditions in agreement with previous studies (e.g. Chamove, 1989). Furthermore, redirected behaviour to other pen mates such as tail biting, ear chewing or belly-nosing was also significantly lower in the animals reared in an enriched environment as previously reported (De Jong et al., 1998; Jensen et al., 2010; Petersen et al., 1995). Moreover, pigs reared in enriched environments spent more time active than pigs reared in barren conditions in agreement with previous results (Beattie et al., 1995; Bolhuis et al., 2005; Bracke and Spoolder, 2008). However, the activity of EEHC was not enhanced, and presented similar levels to HC and CG. A possible hypothesis to explain the difference between EE and EEHC is somehow related with the sedative properties of the herbal compound which can counteract a hypothetical increase of the activity produced by the enriched material. Nevertheless, HC pigs did not differ from CG with regards to activity patterns. Thus, more research is needed to better clarify the potential reduction on the activity modulated by the herbal compounds. In fact, if herbal compounds are able to produce a reduction of the activity in pigs, this reduction must not be confounded by the lack of motivation produced by barren environments or inappropriate enrichment material.

With regards to novel tests, pigs in barren conditions crossed the two meter line from the door to the novel object test quicker than pigs reared with enrichment material when a PVC ball was used. Previous studies have also reported an increased excitability, a quicker approach and longer interactions in front of a novel object in pigs reared in barren conditions compared with enriched conditions (Bracke and Spoolder, 2008; Grandin et al., 1987; Pearce and Paterson, 1993; Stolba and Wood-Gush, 1980). Stolba and Wood-Gush (1980) argued that pigs usually kept in barren environments could present a stronger motivation to explore what could support the higher latency of enriched pigs to cross the two meters line compared to the barren pigs. However, no other significant effect was found for this or the other novelty tests used.

5. Conclusions

The results of the present work highlight a positive effect of rearing pigs in an enriched environment and/or supplementing the diet with a herbal compound containing *Valeriana officinalis* and *Passiflora incarnata*. Environmental enrichment reduced some abnormal behaviours such as stereotypies and redirected behaviour, while increased exploratory behaviour. On the other hand, pigs supplemented with the herbal compound presented less negative interactions and less skin injuries. Furthermore, pigs reared in enriched conditions, supplemented with a herbal compound or with both treatments at the same time presented higher body weight compared to the control pigs. Although no economical analysis was carried out in the present experiment, this better performance when enrichment was provided could have commercial implications (i.e. enhanced profitability) that would be interesting to further investigate. The reduction of redirected behaviours associated with the provision of enrichment may have also commercial implications related to the development of strategies to reduce tail biting and avoid tail docking.

Conflicts of interest

The authors declare no conflict of interest with regards to the contents of the paper.

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